

Algebra II with Trigonometry

Chapter 6.3

Olympiad Solutions (Gemini)

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Solutions

1. Find the sum of all real solutions to the equation $x = 2024\pi \sin(x)$.

Let $f(x) = x - 2024\pi \sin(x)$. Finding the solutions to the given equation is equivalent to finding the roots of $f(x) = 0$.

First, note that the function is odd because

$f(-x) = -x - 2024\pi \sin(-x) = -(x - 2024\pi \sin(x)) = -f(x)$. Because $f(x)$ is odd, any non-zero real roots must occur in symmetric pairs, $(r, -r)$.

Next, we establish that the number of roots is finite. Since $|\sin(x)| \leq 1$ for all real x , any root must satisfy $|x| = |2024\pi \sin(x)| \leq 2024\pi$. Thus, all real roots lie within the bounded interval $[-2024\pi, 2024\pi]$. Since the roots are finite in number and symmetrically distributed around the origin (with $x = 0$ itself being a root), their total sum is exactly 0.

2. Determine the difference between the absolute maximum and absolute minimum values of the function $f(x) = (\sin x + \cos x)^4$ over the real numbers.

We can elegantly simplify the function by expanding the inner square:

$$f(x) = ((\sin x + \cos x)^2)^2 = (\sin^2 x + \cos^2 x + 2 \sin x \cos x)^2 = (1 + \sin 2x)^2$$

Because the range of $\sin 2x$ over the real numbers is $[-1, 1]$, the expression $1 + \sin 2x$ takes all values in the interval $[0, 2]$.

Squaring this interval, the minimum value of $f(x)$ is $0^2 = 0$, and the maximum value is $2^2 = 4$. Therefore, the difference between the absolute maximum and the absolute minimum is $4 - 0 = 4$.

3. How many real numbers x satisfy the equation

$$\sin(x) + \cos(x\sqrt{2}) = 2?$$

For all real x , the maximum values of both $\sin(x)$ and $\cos(x\sqrt{2})$ are 1. Their sum can equal 2 if and only if both terms simultaneously achieve their maximum: 1) $\sin(x) = 1 \implies x = \frac{\pi}{2} + 2k\pi$ for some integer k . 2) $\cos(x\sqrt{2}) = 1 \implies x\sqrt{2} = 2m\pi$ for some integer m .

Equating the two expressions for x , we have:

$$\frac{\pi}{2} + 2k\pi = \frac{2m\pi}{\sqrt{2}}$$

Dividing both sides by π yields:

$$k + \frac{1}{4} = \frac{m}{\sqrt{2}}$$

Because k is an integer, the left side is always a non-zero rational number. Conversely, $m/\sqrt{2}$ is irrational for any non-zero integer m , and zero if $m = 0$. A non-zero rational number can never equal an irrational number or zero, meaning no such integers k and m exist.

Hence, there are 0 real numbers that satisfy the equation.

4. Find the sum of the absolute maximum and absolute minimum values of the function $f(x) = \cos(2x) - 4\sin(x)$.

Using the double-angle identity $\cos(2x) = 1 - 2\sin^2(x)$, we can rewrite the function entirely in terms of $\sin(x)$:

$$f(x) = 1 - 2\sin^2(x) - 4\sin(x)$$

Let $u = \sin(x)$, where $u \in [-1, 1]$. The function becomes a quadratic in u :

$$g(u) = -2u^2 - 4u + 1 = -2(u^2 + 2u + 1) + 3 = -2(u + 1)^2 + 3$$

This parabola opens downward and has its vertex at $u = -1$. Because $u = -1$ is within our interval $[-1, 1]$, the absolute maximum is obtained at the vertex:

$$g(-1) = 3$$

The absolute minimum occurs at the endpoint furthest from the vertex, which is $u = 1$:

$$g(1) = -2(1 + 1)^2 + 3 = -2(4) + 3 = -5$$

The sum of the absolute maximum and absolute minimum values is $3 + (-5) = -2$.

5. The function $v(t) = \sin\left(\frac{t}{a}\right) + \cos\left(\frac{t}{b}\right)$, where a and b are relatively prime positive integers, models the combined sound wave of two tuning forks. If the fundamental period of $v(t)$ is 120π , find the minimum possible value of the sum $a + b$.

The individual period of $\sin(t/a)$ is $2\pi a$, and the individual period of $\cos(t/b)$ is $2\pi b$. The fundamental period of their sum is the least common multiple (LCM) of their individual periods:

$$\text{Period} = \text{lcm}(2\pi a, 2\pi b) = 2\pi \cdot \text{lcm}(a, b)$$

We are given that the period is 120π , which means $\text{lcm}(a, b) = 60$. Since a and b are relatively prime ($\text{gcd}(a, b) = 1$), their least common multiple is simply their product. Thus, $ab = 60$.

We look for pairs of relatively prime positive integers (a, b) whose product is 60. The possible pairs (ignoring order, since we only need the sum) are:

- $(1, 60) \implies 1 + 60 = 61$
- $(3, 20) \implies 3 + 20 = 23$
- $(4, 15) \implies 4 + 15 = 19$
- $(5, 12) \implies 5 + 12 = 17$

The minimum possible value of the sum $a + b$ is 17.

6. Determine the exact amplitude of the combined sinusoidal wave given by $y = \sin(x) + \sin\left(x + \frac{\pi}{3}\right) + \sin\left(x + \frac{2\pi}{3}\right)$.

We can gracefully evaluate this using the sum-to-product trigonometric identity on the first and third terms:

$$\sin(\alpha) + \sin(\beta) = 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$$

Applying this to $\sin(x)$ and $\sin\left(x + \frac{2\pi}{3}\right)$:

$$\sin(x) + \sin\left(x + \frac{2\pi}{3}\right) = 2 \sin\left(x + \frac{\pi}{3}\right) \cos\left(-\frac{\pi}{3}\right)$$

Since $\cos\left(-\frac{\pi}{3}\right) = \frac{1}{2}$, this beautifully collapses to:

$$2 \sin\left(x + \frac{\pi}{3}\right) \cdot \frac{1}{2} = \sin\left(x + \frac{\pi}{3}\right)$$

Now, substituting this back into the original wave equation:

$$y = \sin\left(x + \frac{\pi}{3}\right) + \sin\left(x + \frac{\pi}{3}\right) = 2 \sin\left(x + \frac{\pi}{3}\right)$$

The exact amplitude of the combined sinusoidal wave is 2.